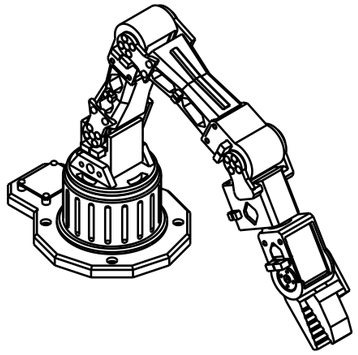
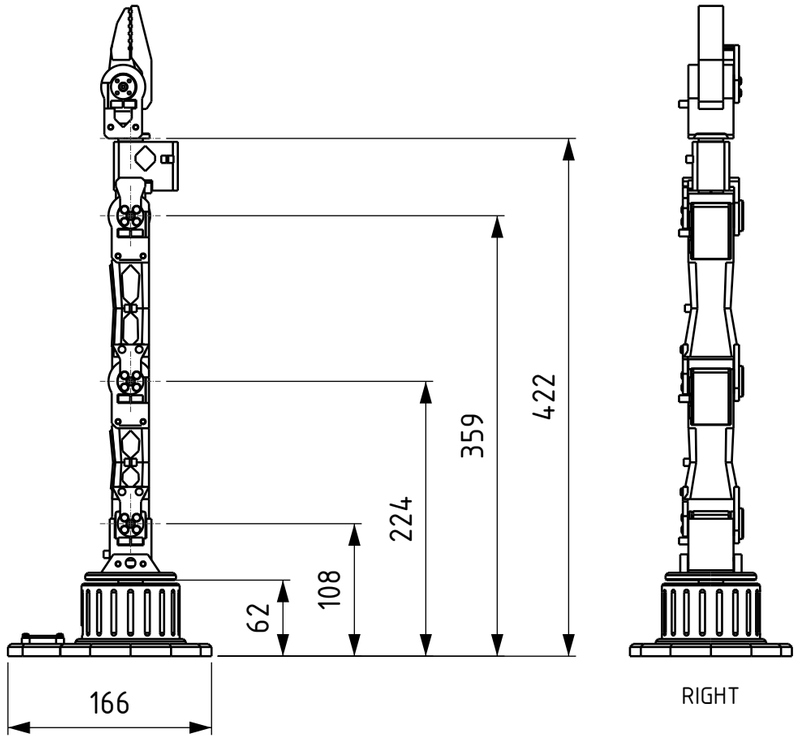


TOP



ISOMETRIC (NOT TO SCALE)



FRONT

RIGHT

PRINTED PARTS

PART NO.	NAME	QTY	SET
TAL-101	Base	1	A
TAL-102	Shoulder (turntable + cradle)	1	A
TAL-103	Upper arm	1	A
TAL-104	Forearm	1	B
TAL-105	Wrist	1	B
TAL-106	Gripper body	1	B
TAL-107	Moving jaw	1	B
TAL-108	Hub cap	3	A
TAL-110	Idler plate	2	B
TAL-111	Servo fit gauge (print first)	1	A

BOUGHT PARTS

ITEM	QTY
Feetech STS3215 servo (12 V)	6
Waveshare Bus Servo Adapter (A)	1
M3 x 6 button head (output and passive horns)	36
M3 x 10 button head (idler plates)	4
M2 x 5 self-tapping (servo cases)	12
M2 x 6 self-tapping (servo cases)	8
M2.5 x 6 self-tapping (board)	4
M4 bolts to the table (user's)	4
3-pin bus servo cables (with servos)	6

JOINT LAYOUT

1. Pan axis vertical through the base centre; pan horn face 62 above the table. Plinth 132 across corners.
2. Shoulder axis (along Y) 46 above the pan horn face. Upper arm 116, forearm 135 (axis to axis). Wrist flex axis to roll horn face 63.
3. Front view: zero pose (all joints 0). Isometric: 'ready' pose.
4. Assembly, each joint: plug a 3-pin cable into the servo; push the servo into its part along its entry path (part sheet notes); fit its M2 case screws; fit both horns (output horn centre screw; the passive horn sits loose on its boss); fit the next link with 4X M3 × 6 into each horn.
5. Print TAL-111 first and check the servo fit before printing the arm. Each part sheet gives the print orientation; print-ready 3MF files are in cad/print.

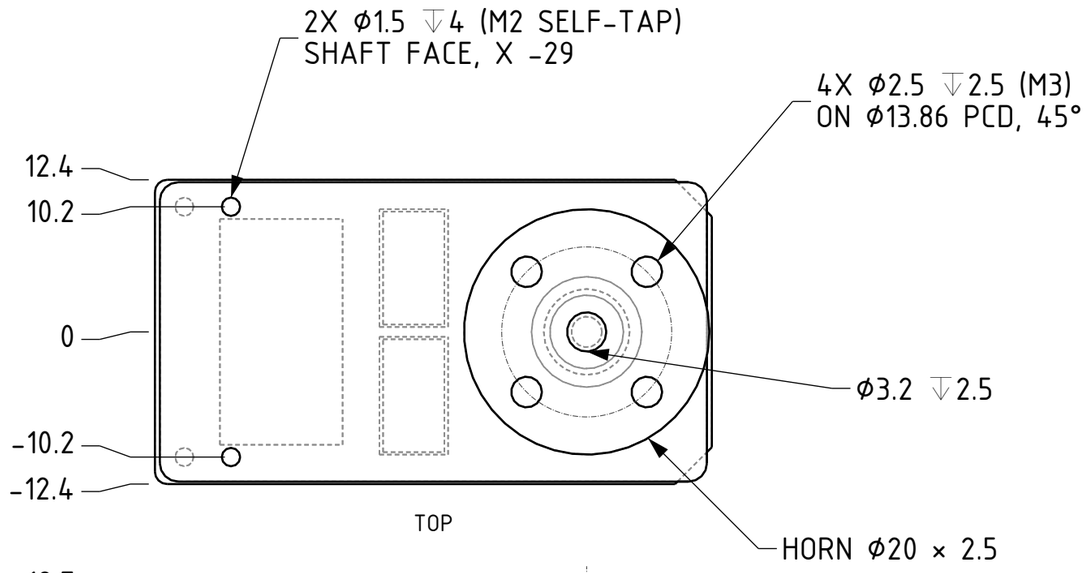
NOTES

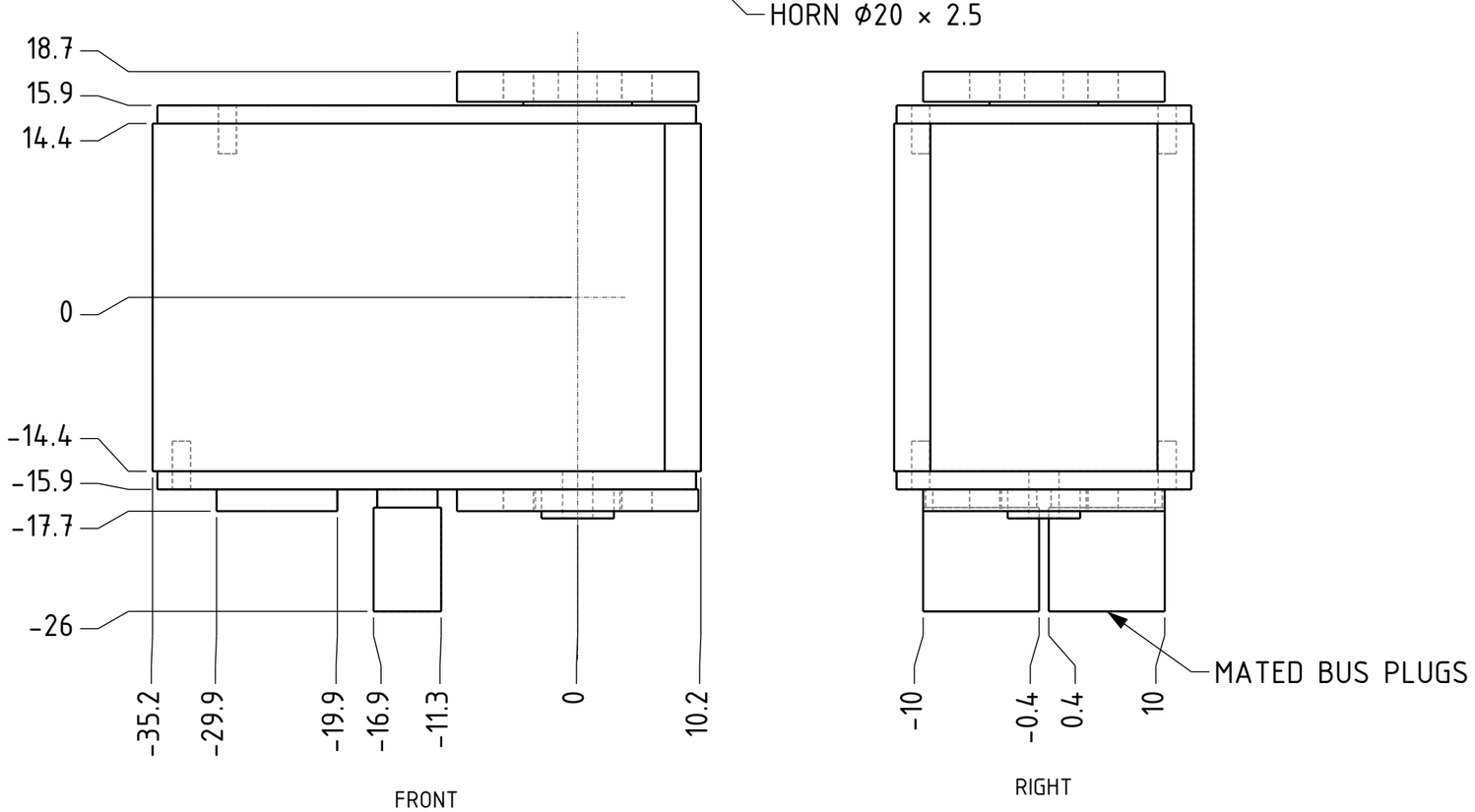
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part sheets give each part in its own frame; this sheet shows where those frames sit.

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B

PROJECT TALOS 5-axis arm	EST. MASS -
PART General assembly and parts list	SCALE 1:5
PART NO. TAL-000	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-06
MATERIAL -	SHEET 1 of 21





SERVO (servo frame)

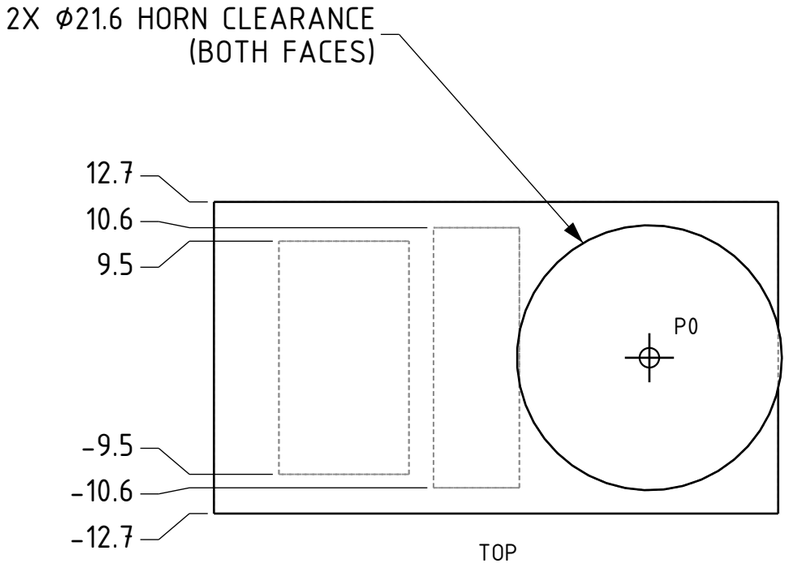
1. Body 45.4 × 24.8 × 28.8, x −35.2 to 10.2; +X end corners C3, −X end corners R1.
2. Caps 44.6 × 24.4 × 1.5, corners R1.5, centred on x −12.5, on both faces.
3. Back pad (connector face) x −29.9 to −19.9, y ±9.2, 1.8 high.
4. Bus headers x −16.6 to −11.6, y 0.6 to 9.8 and −9.8 to −0.6; mated plugs to z −26.
5. Output horn ⌀20 × 2.5 (z 16.2 to 18.7): 4X ⌀2.5 M3 on ⌀13.86 PCD at 45°, centre ⌀3.2.
6. Passive horn ⌀20 × 1.8 on the connector face, centre ⌀7.
7. Case holes ⌀1.5 ↧4 (M2 self-tap): shaft face at (−29, ±10.2), connector face at (−32.8, ±10.2).

NOTES

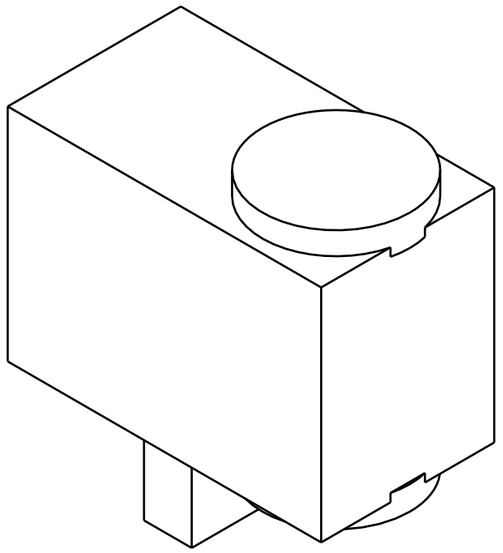
- Bought part: dimensions for reference, measured from the SO-ARM100 STS3215 model and Feetech's drawing. Check the connector position and case-hole depth on a real servo.
- Sheet 2 gives pocket P, the clearance envelope that every servo part cuts.

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART STS3215 servo (reference) (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM 6	DATE 2026-11-06
MATERIAL bought part	SHEET 2 of 21



TOP



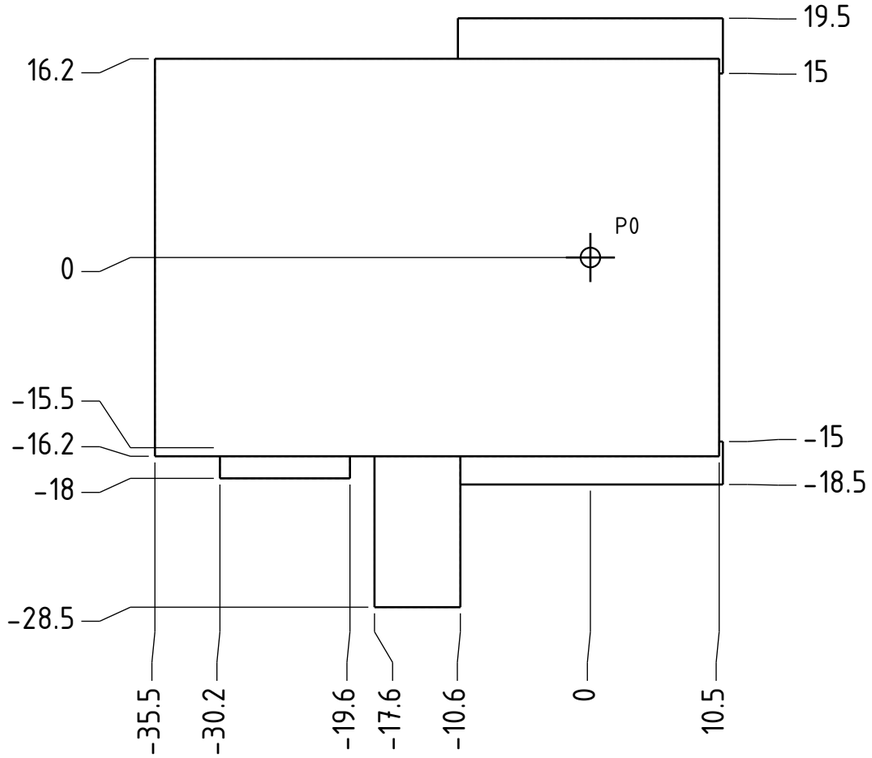
ISOMETRIC (NOT TO SCALE)

POCKET P (servo frame; 0.3 clearance included)

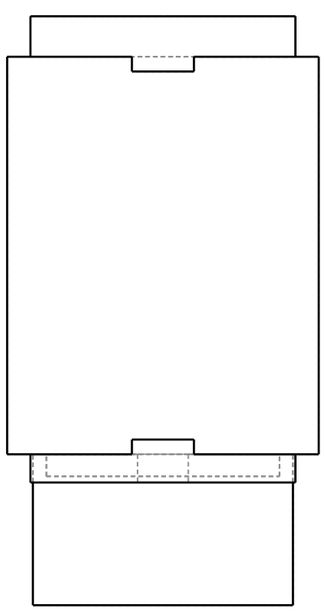
1. Main box x −35.5 to 10.5, y ±12.7, z ±16.2.
2. Back pad box x −30.2 to −19.6, y ±9.5, z −18.0 to −15.5.
3. Plug bay box x −17.6 to −10.6, y ±10.6, z −28.5 to −15.5 (mated plugs + cable bend).
4. Idler clearance cylinder ⌀21.6 on the shaft axis, z −18.5 to −15.0.
5. Horn clearance cylinder ⌀21.6 on the shaft axis, z 15.0 to 19.5.

NOTES

- Each part sheet locates P0 and gives the servo axes in the part frame; cut the union of the five bodies.
- All dimensions mm. Third-angle projection. Do not scale the drawing.



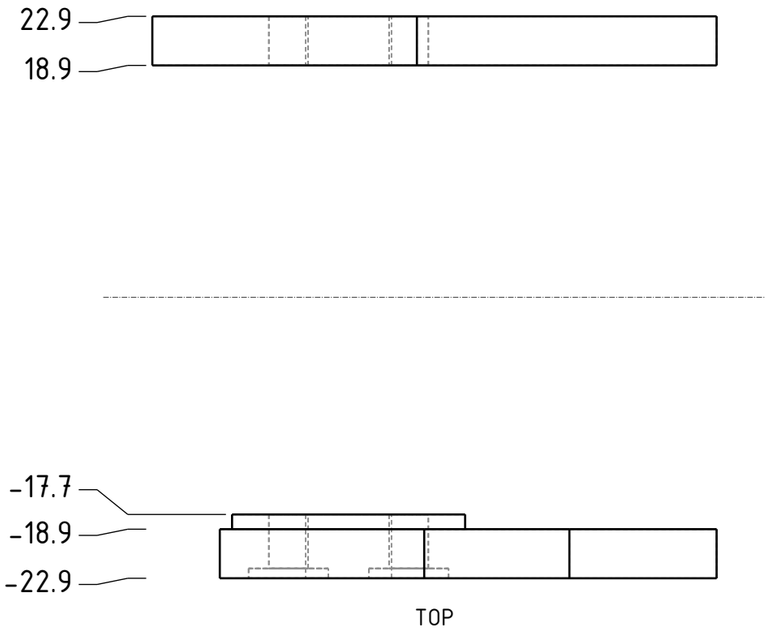
FRONT



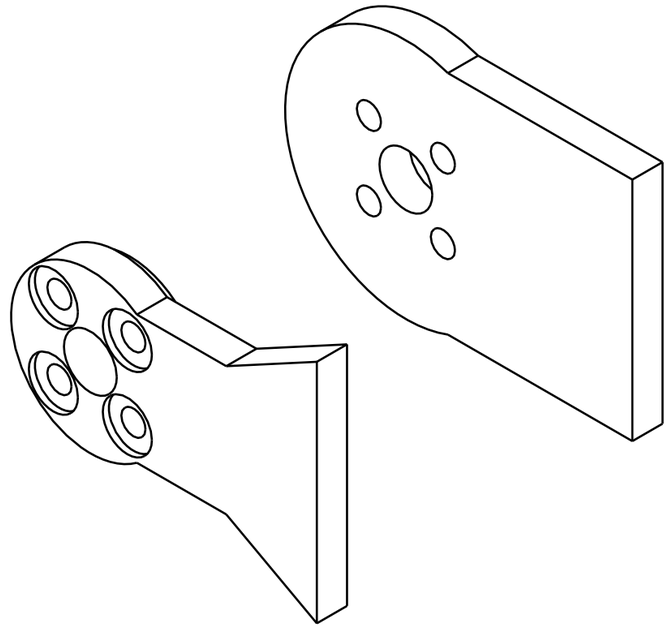
RIGHT

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Servo pocket P (cut into parts) (part sheet 2 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-06
MATERIAL -	SHEET 3 of 21



TOP



ISOMETRIC (NOT TO SCALE)

IDLER NECK POINTS (X, Z), MIRRORED IN Z

PT	X	Z
N1	0	±10.5
N2	6	±8.5
N3	0.6R = 18	±8.5
N4	R = 30	±H = ±15

CLEVIS (every pitch joint)

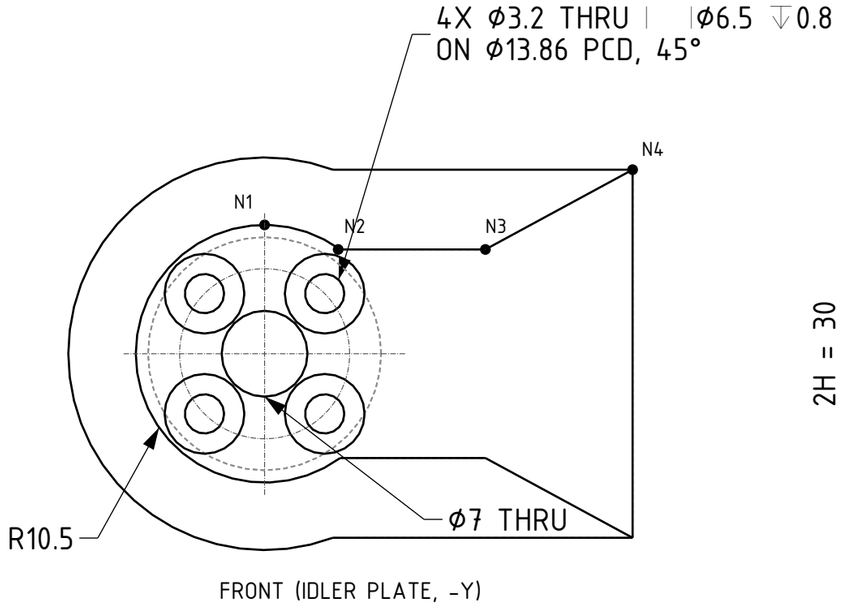
Drawn for the arm links: reach R = 30, half-height H = 15. The wrist uses R = 24, H = 12.

Not printed on its own: the plates are modelled into the links. On TAL-103 and TAL-104 the idler plate is the separate part TAL-110; on the wrist (TAL-105) both plates are wrist body.

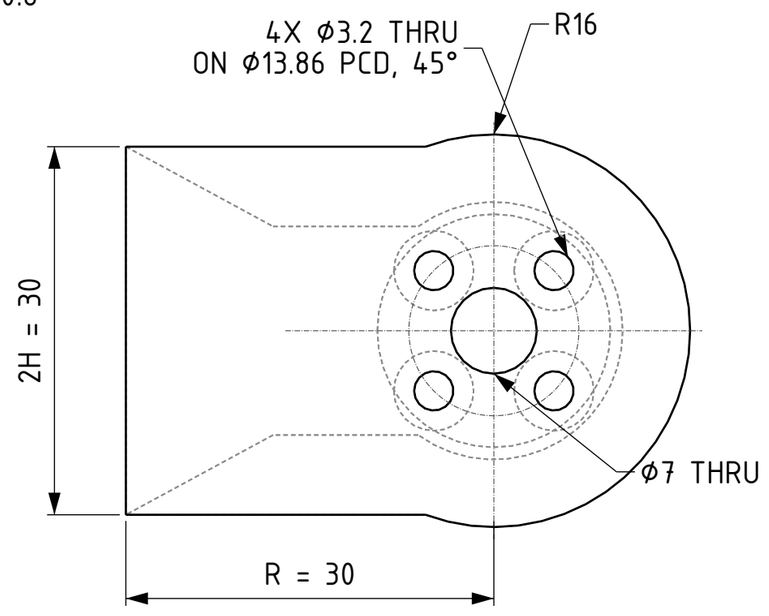
1. Output plate (+Y), y 18.9 to 22.9: circle R16 at the origin + rectangle x 0 to R, z ±H.
2. Output plate holes, through: Ø7 centre (horn spline screw) + 4X Ø3.2 on Ø13.86 PCD at 45°.
3. Idler plate (-Y), y -22.9 to -18.9: circle R10.5 + neck polygon N1 to N4 (mirrored). The neck clears the parent servo's bus plugs.
4. Idler plate pad Ø19 on the inner face, y -18.9 to -17.7, bearing on the passive horn; centre Ø7 through (clears the idler boss); 4X Ø3.2 on Ø13.86 PCD at 45°, counterbore Ø6.5 0.8 deep from outside (4X M3 × 6 into the passive horn, which turns on the boss).
5. Clevis gap in the link body: remove a cylinder R21 about Y between y ±18.9, and everything at x < 0 between those planes (dimensioned on each link's section).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.



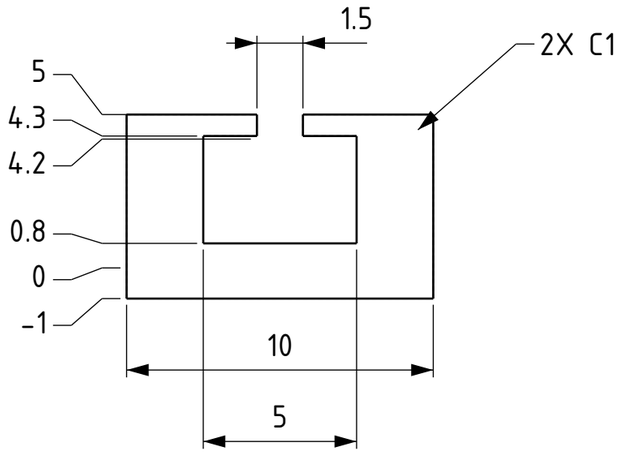
FRONT (IDLER PLATE, -Y)



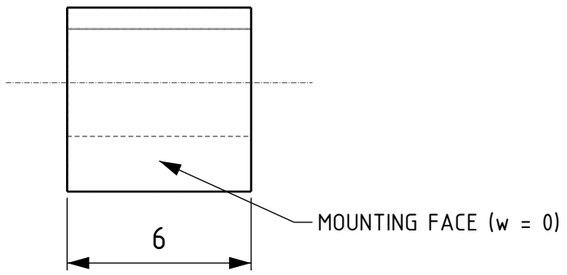
REAR (OUTPUT PLATE, +Y)

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clevis (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 4 of 21



CABLE CLIP: END VIEW



CABLE CLIP: SIDE VIEW

SEAT S (FROM THE SERVO MID-PLANE)

PART	S
arm shells, gripper	17.3
shoulder	18.5
base	18.4
wrist	18.0

COMMON FEATURES

CABLE CLIP, one bus cable. Local u along the cable, v across, w out of the mounting face.

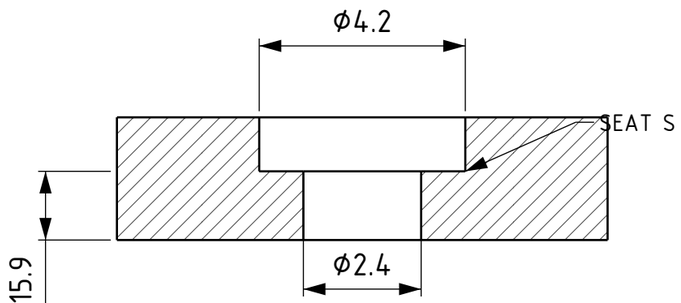
1. Block u ±3, v ±5, w -1 to 5 (sunk 1 into the face).
2. Tunnel through along u: v ±2.5, w 0.8 to 4.3. Slit v ±0.75 from w 4.2 out through the top.
3. Chamfer C1 on the two outer long edges.

CASE SCREW HOLE (M2 self-tap into the servo case): ⌀2.4 from the servo face to the seat S, then ⌀4.2 counterbore from S outward through the part.

HEX WINDOW H(xa, xb, zc, k): hexagon (xa, zc) (xa+k, zc-k) (xb-k, zc-k) (xb, zc) (xb-k, zc+k) (xa+k, zc+k), cut through the part along Y. Drawn for H(0, 40, 0, 10).

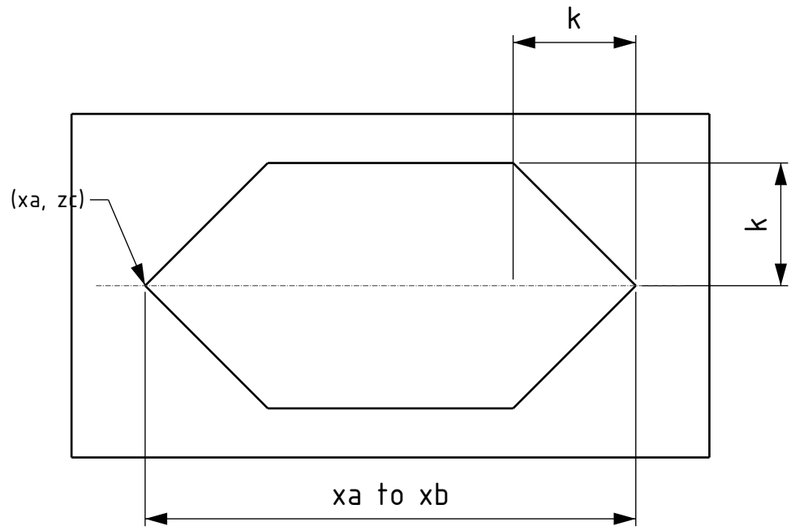
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.



CASE SCREW HOLE (SECTION) SCALE 8:1

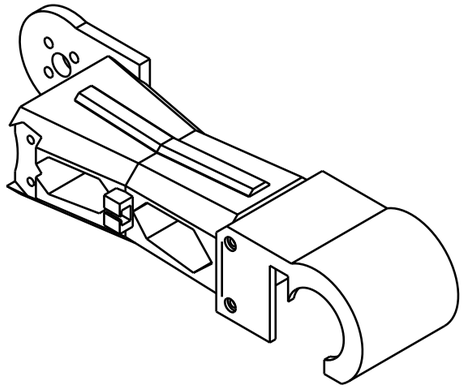
SERVO FACE (BOTTOM) IS 15.9
FROM THE SERVO MID-PLANE



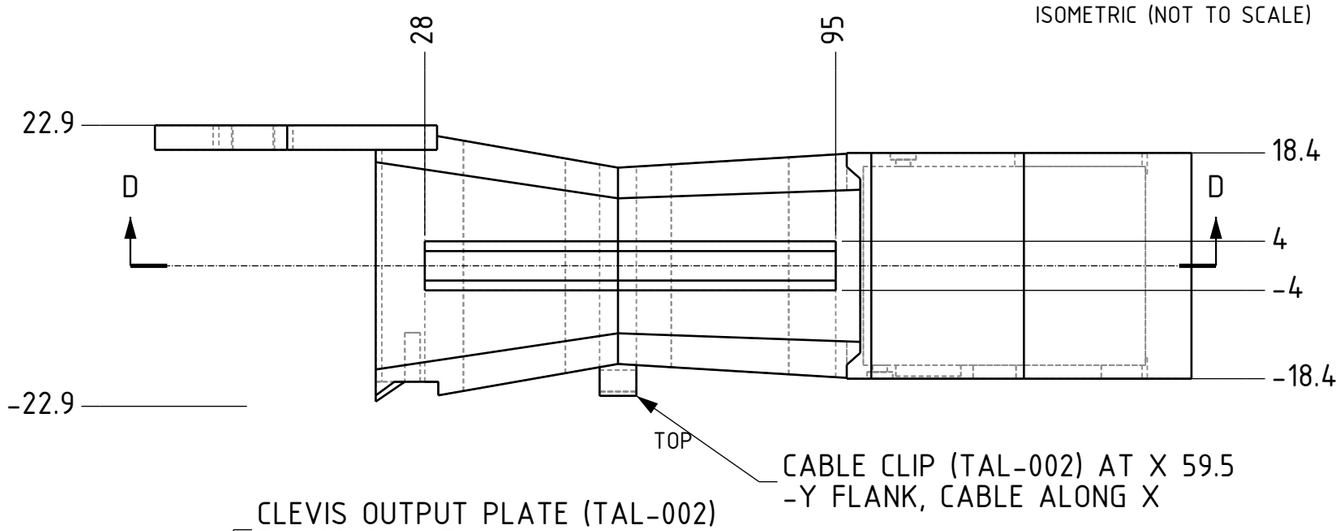
HEX WINDOW H(xa, xb, zc, k) SCALE 2:1

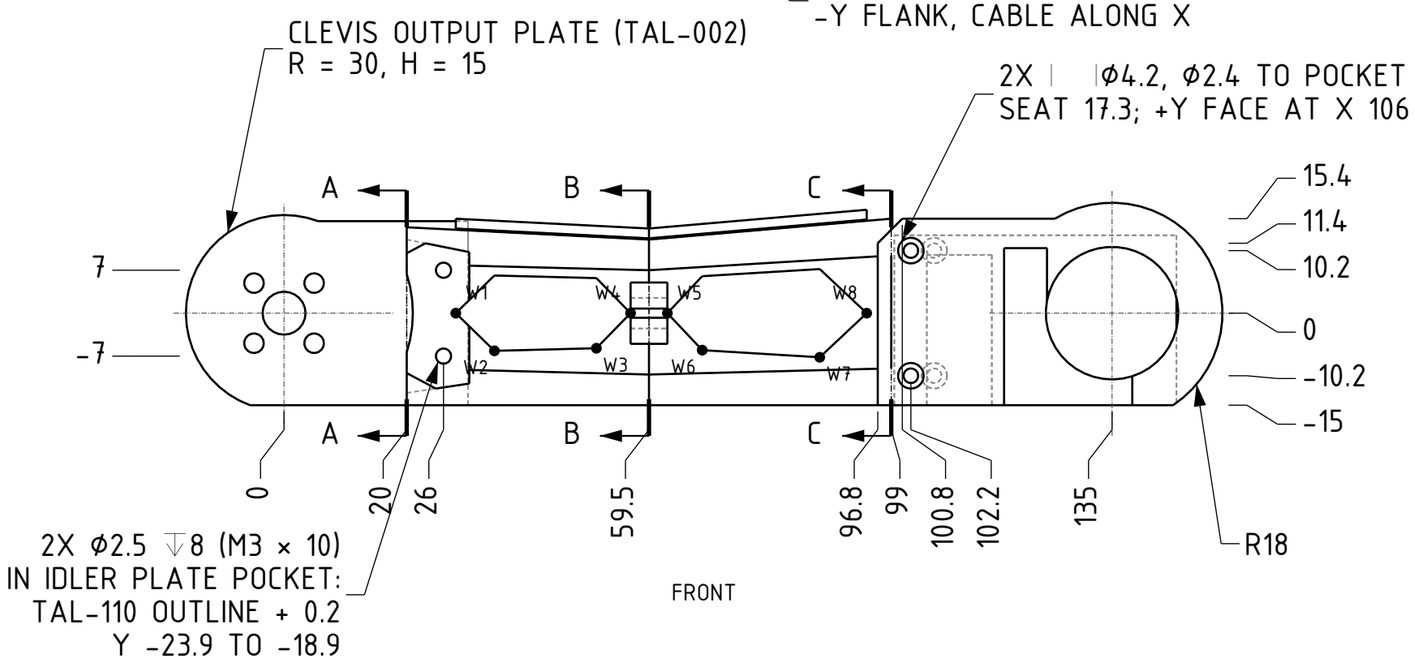
THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clip, screw hole, window (part sheet 2 of 2)	SCALE 5:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 5 of 21



ISOMETRIC (NOT TO SCALE)





WINDOW POINTS (X, Z), THROUGH Y

PT	X	Z
W1	28	0
W2	34.3	±6.14
W3	50.92	±5.72
W4	56.5	0
W5	62.5	0
W6	68.18	±6.03
W7	87.34	±7.19
W8	95	0

FEATURES (in order)

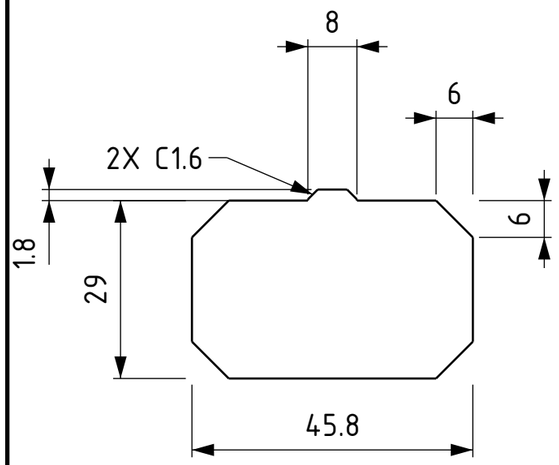
1. Clevis output plate only (TAL-002, +Y plate), R = 30, H = 15. The idler plate is the separate part TAL-110.
2. Beam: ruled loft of the profiles A, B, C (sheet 2) at x 20, 59.5, 99. Every profile sits on z −15, so the underside is flat; the waist dips on top only.
3. Servo shell: XZ polygon (96.8, −15) (135, −15) (135, 15.4) (100.8, 15.4) (96.8, 11.4), extruded y ±18.4; plus a cylinder R18 about Y at x 135.
4. Flat underside: cut everything below z −15 (trims the clevis hub and the R18 cylinder).
5. Spine: ruled loft of the ridge profiles A, B, C, trimmed to x 28 to 95.
6. Windows: polygons W1 to W4 and W5 to W8 (mirrored in Z), cut through Y.
7. Cable clip (TAL-002) on the −Y flank at x 59.5, cable along X.
8. Clevis gap cut (TAL-002): R21 about Y between y ±18.9, plus x < 0 between those planes.
9. Pocket P (sheet TAL-001) with its servo frame origin P0 at (135, 0, 0): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along −Z.
10. Servo entry from below: cut box x 99.5 to 145.5, y ±16.2, z below 12.7 (opens the floor under the whole servo). In the −Y wall: back pad channel x 104.8 to 115.4, y −18 to −16.2, z below −9.5; notch through the wall x 117.4 to 138.3, z below 0 (joins the plug window and the idler hole).
11. Case screw holes (TAL-002, seat 17.3): +Y face at x 106, z ±10.2; −Y face at x 102.2, z ±10.2.
12. Idler plate pocket: the TAL-110 outline (circle R10.5 + neck N1 to N4) grown 0.2, cut from y −23.9 to −18.9.
13. Idler plate screw holes: 2X ⌀2.5 at x 26, z ±7, from y −18.9 8 deep (M3 × 10).

NOTES

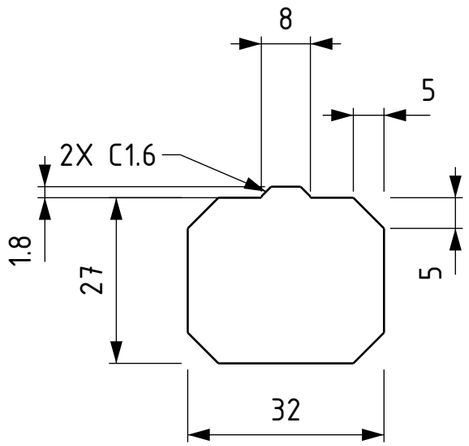
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/−0.2 unless noted; servo pocket clearances are already included.
- Fit the wrist flex servo from below, shaft to +Y, before its horns and plugs; then 4X M2 × 5 case screws.
- The wrist flex servo's bus plugs come out through the −Y shell wall.
- Print: flat underside down; no support (servo-opening roof and window tops bridge). The flat underside is z −15.

THIRD ANGLE PROJECTION

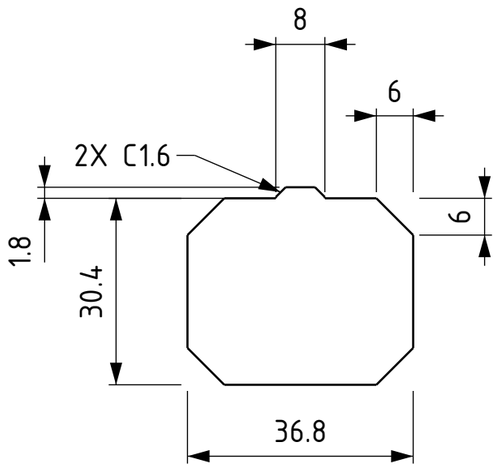
WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS 52 g
PART Forearm (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-104	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 12 of 21



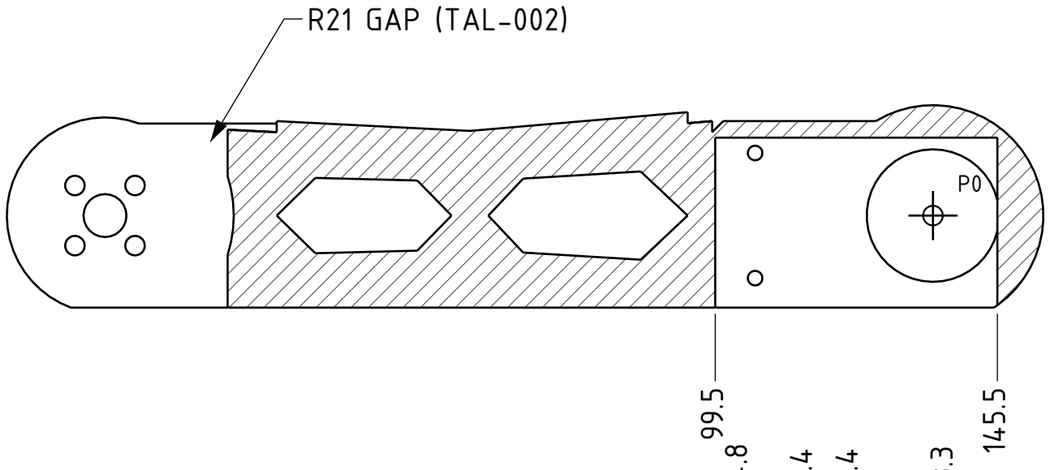
PROFILE A-A (X 20)



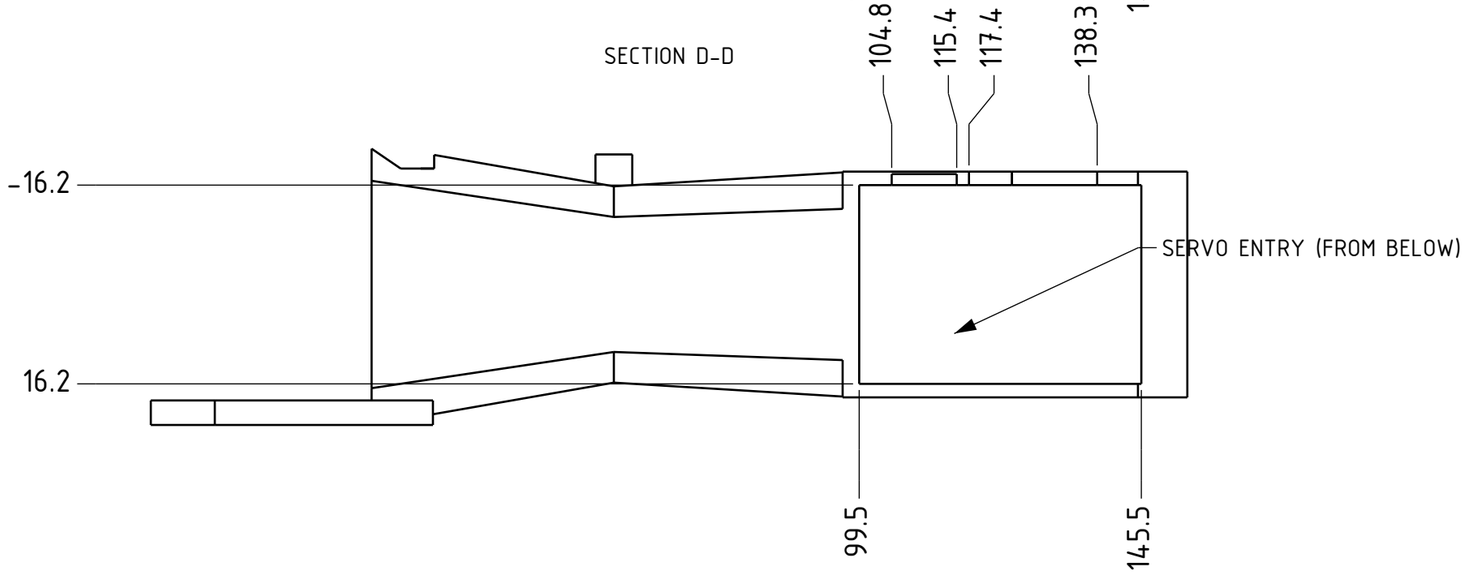
PROFILE B-B (X 59.5)



PROFILE C-C (X 99)



SECTION D-D



BOTTOM

PROFILES AND SECTION

Profiles A, B, C: beam octagon (corners chamfered) with the spine ridge on top, in the YZ plane at the loft stations; cutting lines on sheet 1. Each sits on z −15 (flat underside) with its top at the height given.

Ridge: 8 wide, 4.8 across the top, 1.8 above the beam, 45° bevels; its base runs 3 into the beam.

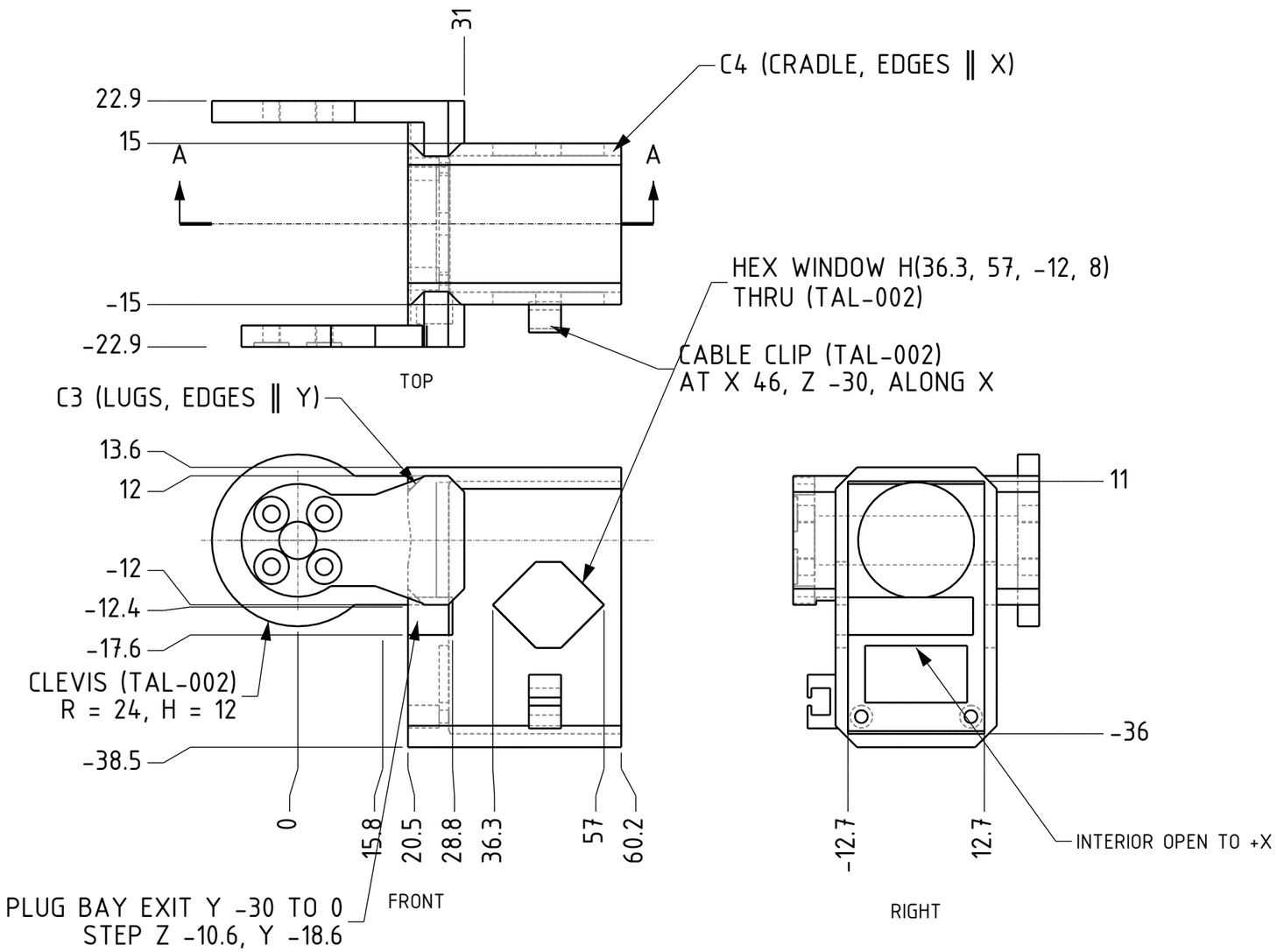
Section D-D: XZ plane, looking toward +Y. P0 = pocket P servo frame origin.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Forearm (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-104	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 13 of 21



FEATURES (in order)

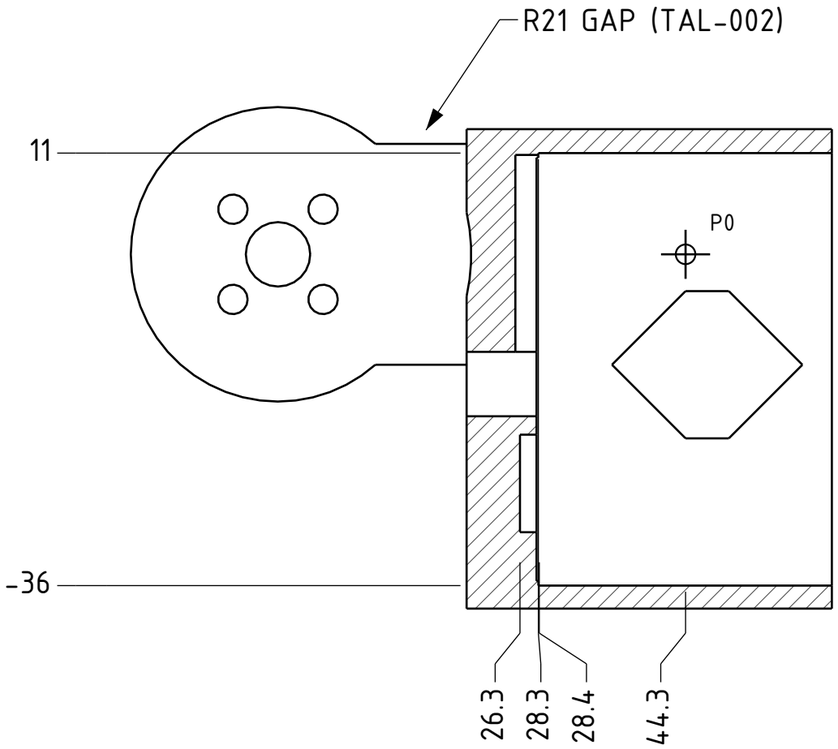
1. Clevis (TAL-002), R = 24, H = 12.
2. Bridge lugs: box x 20.5 to 31, y ±22.9, z ±12; C3 on the edges parallel to Y.
3. Cradle: box x 20.5 to 60.2, y ±15, z -38.5 to 13.6; C4 on the edges parallel to X.
4. Cable clip (TAL-002) on the -Y face: centre (46, -15.2, -30), cable along X.
5. Clevis gap cut (TAL-002).
6. Pocket P (sheet TAL-001) with its servo frame origin P0 at (44.3, 0, 0): servo +Z (shaft) along +X, servo +X along +Z (body hangs down), servo +Y along -Y.
7. Cradle interior: cut box x 28.3 to the +X end, y ±12.7, z -36 to 11.
8. Plug bay exit to -Y: cut boxes x 15.8 to 28.8: y -18.6 to 0, z -17.6 to -10.6; and y -30 to 0, z -17.6 to -12.4.
9. Side windows: hex window H(36.3, 57, -12, 8) through Y (TAL-002).
10. Case screw holes (TAL-002, seat 18.0), connector face only, from the back wall: (y, z) = (±10.2, -32.8).

NOTES

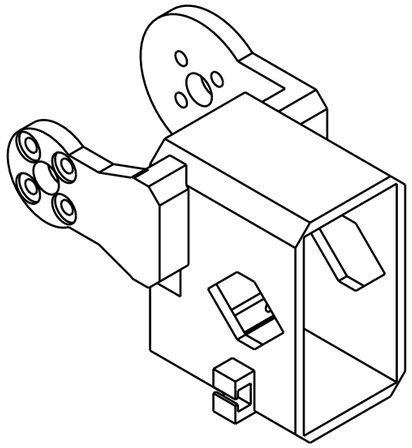
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Fit the roll servo's case screws before the gripper goes on.
- Print: underside down as drawn; support only under the plug bay roof (servo-opening roof bridges).

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS 29 g
PART Wrist (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-105	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 14 of 21



SECTION A-A



ISOMETRIC (NOT TO SCALE)

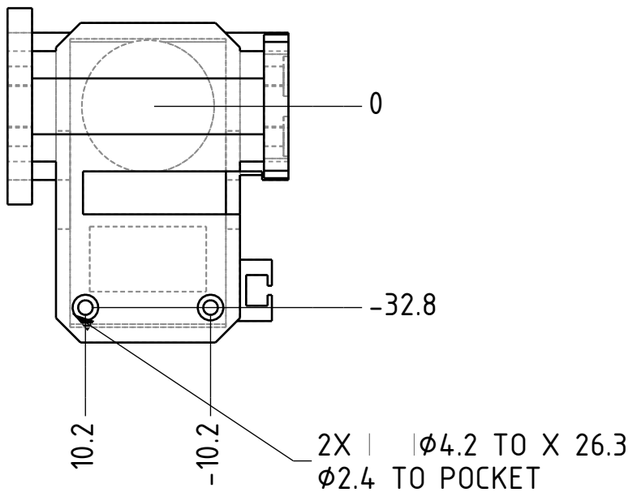
SECTION A-A AND LEFT VIEW

A-A: XZ plane (cutting line on sheet 1), looking toward +Y. P0 = pocket P servo frame origin.

Left view: the cradle back wall, seen from -X between the clevis plates.

NOTES

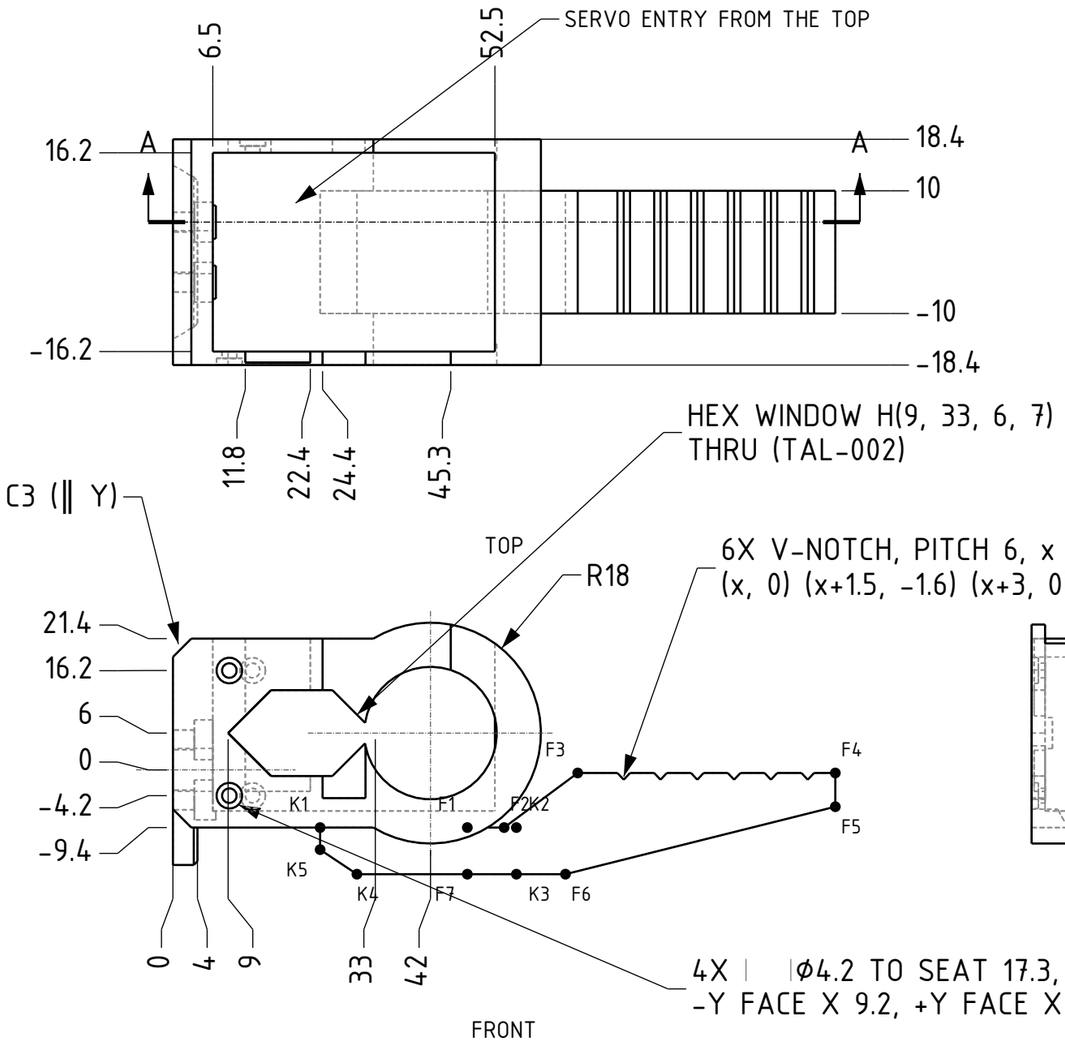
- All dimensions mm. Third-angle projection. Do not scale the drawing.



LEFT SCALE 1:1

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Wrist (part sheet 2 of 2)	SCALE 3:2
PART NO. TAL-105	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 15 of 21



KEEL K AND FINGER F PROFILES (X, Z), Y ±10

PT	X	Z
K1	24	-9.4
K2	56	-9.4
K3	56	-17
K4	30	-17
K5	24	-13
F1	48	-9.4
F2	54	-9.4
F3	66	-0.5
F4	108	-0.5
F5	108	-6
F6	64	-17
F7	48	-17

FEATURES (in order)

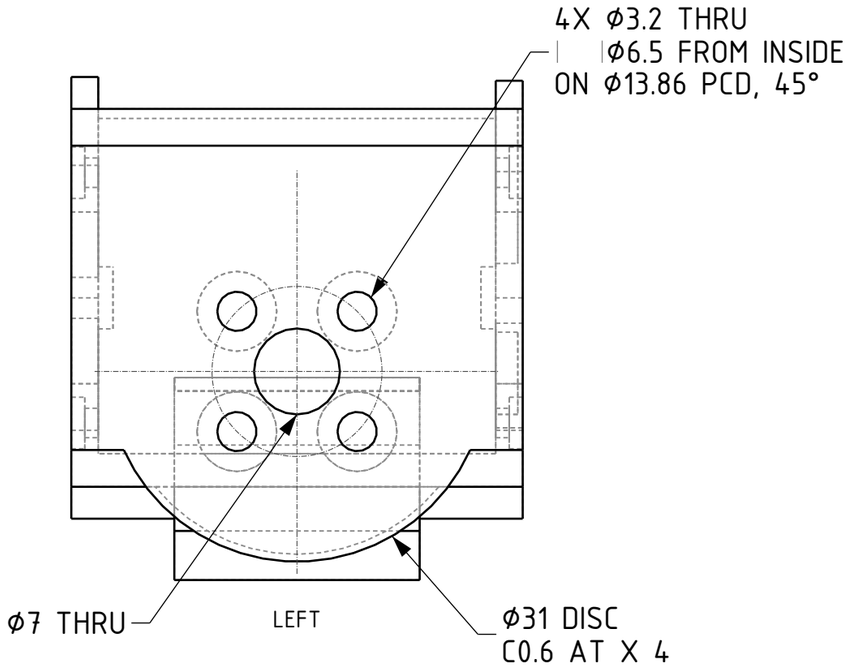
1. Disc ⌀31 along X, x 0 to 4; C0.6 on the front edge.
2. Housing: box x 0 to 42, y ±18.4, z -9.4 to 21.4; C3 on the edges parallel to Y; plus a cylinder R18 about Y at (x, z) = (42, 6).
3. Keel K1 to K5 and finger F1 to F7, each extruded y ±10.
4. Pocket P (sheet TAL-001) with its servo frame origin P0 at (42, 0, 6): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along -Z.
5. Servo entry from the top: cut box x 6.5 to 52.5, y ±16.2, z -6.7 up (opens the top over the whole servo). In the -Y wall: back pad channel x 11.8 to 22.4, y -18 to -16.2, z -3.5 up; notch through the wall x 24.4 to 45.3, z 6 up (joins the plug window and the idler hole).
6. Side windows: hex window H(9, 33, 6, 7) through Y (TAL-002).
7. Grip notches: 6X triangle (x, 0) (x+1.5, -1.6) (x+3, 0), x = 72, 78 … 102, through Y.
8. Roll horn (sheet 2): ⌀7 centre, x -1 to 7. 4X ⌀3.2 on ⌀13.86 PCD at 45°, x -1 to 3.5, counterbore ⌀6.5 from x 3.5 to 7 (M3 × 6 from inside the housing).
9. Case screw holes (TAL-002, seat 17.3): +Y face at x 13, z -4.2 and 16.2; -Y face at x 9.2, z -4.2 and 16.2.

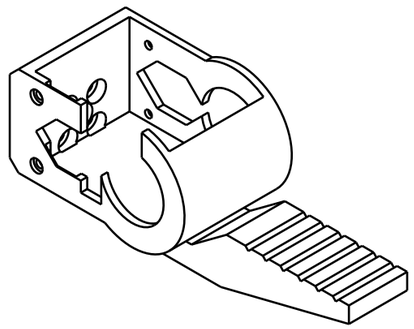
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Fit the jaw servo from the top, shaft to +Y, before its horns and plugs; then 4X M2 × 5 case screws.
- The jaw servo's bus plugs come out through the -Y housing wall.
- Print: standing on the roll-horn disc; no support (one 32 mm bridge at the front of the servo opening); use a brim, the part is 108 tall.

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS 29 g
PART Gripper body (with fixed jaw) (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-106	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 16 of 21





ISOMETRIC (NOT TO SCALE)

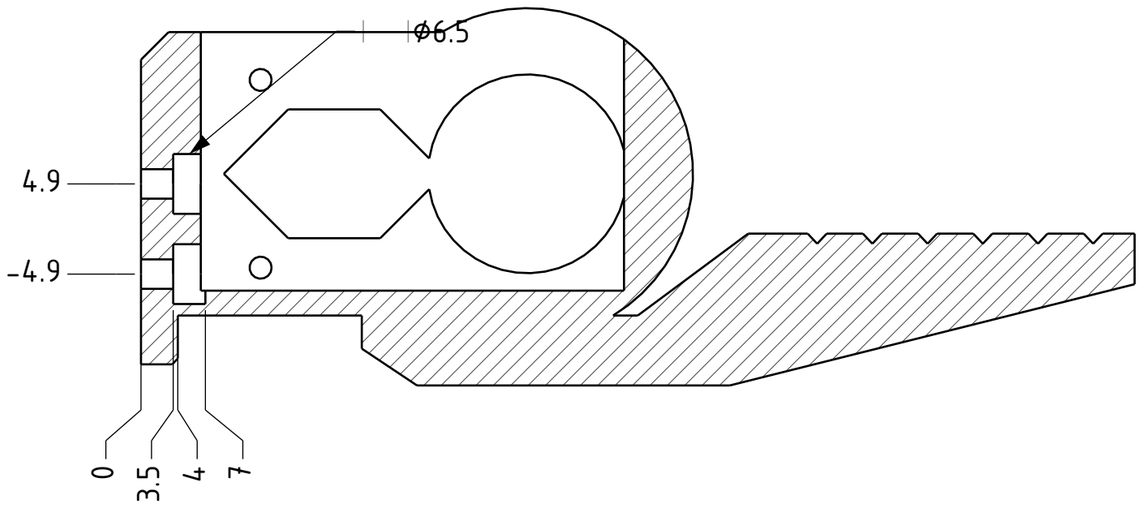
LEFT VIEW AND SECTION A-A

Left view: the disc face (roll horn side), seen from -X.

A-A: plane y 4.9 through two roll horn screws (cutting line on sheet 1, top view), looking toward +Y.

NOTES

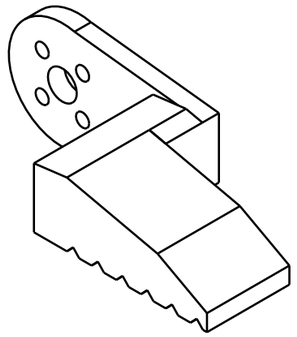
- All dimensions mm. Third-angle projection. Do not scale the drawing.



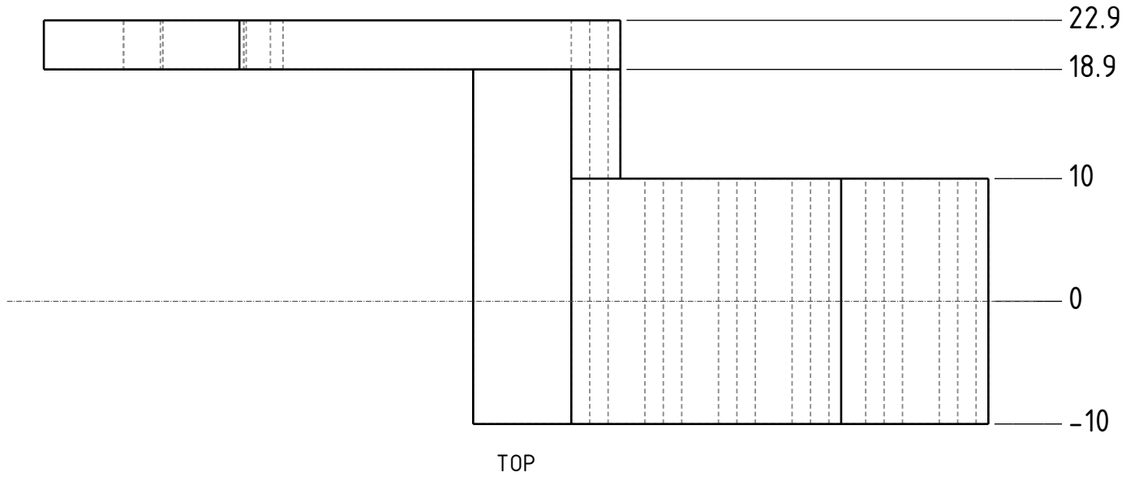
SECTION A-A SCALE 3:2

THIRD ANGLE PROJECTION

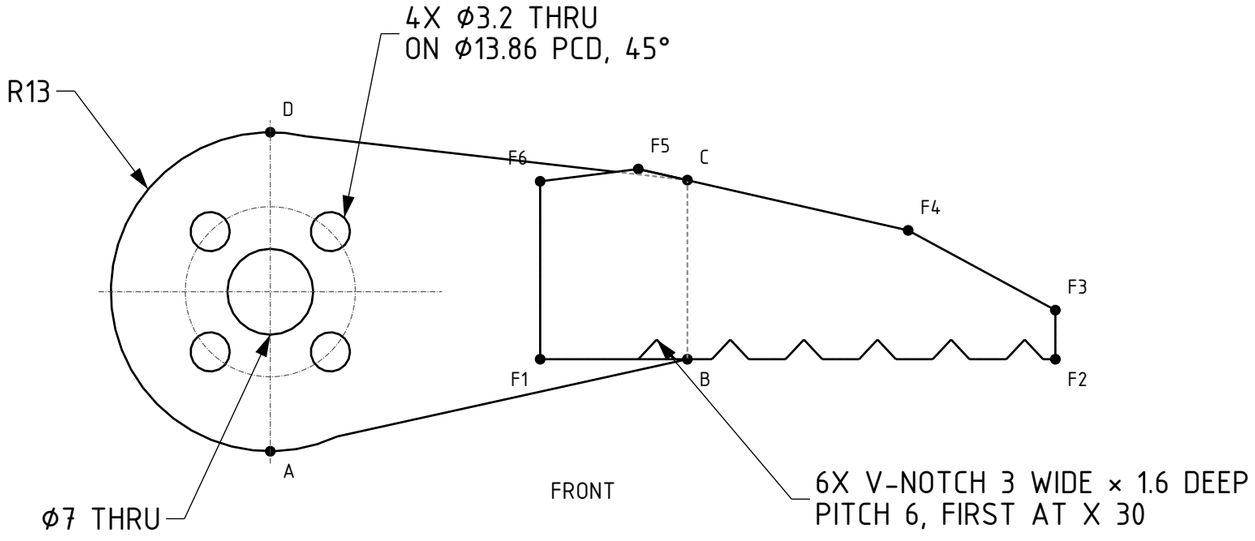
WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Gripper body (with fixed jaw) (part sheet 2 of 2)	SCALE 2:1
PART NO. TAL-106	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 17 of 21



ISOMETRIC (NOT TO SCALE)



TOP



FRONT

PROFILE POINTS (X, Z)

PT	X	Z	PROFILE
A	0	-13	plate
B	34	-5.5	plate, knuckle
C	34	9.1	plate, knuckle
D	0	13	plate
F1	22	-5.5	finger, knuckle
F2	64	-5.5	finger
F3	64	-1.5	finger
F4	52	5	finger
F5	30	10	finger, knuckle
F6	22	9	finger, knuckle

FEATURES (in order)

1. Plate, y 18.9 to 22.9: circle R13 at the origin + polygon A B C D.
2. Finger, y -10 to 10: polygon F1 F2 F3 F4 F5 F6.
3. Knuckle, y 10 to 18.9: polygon F1 B C F5 F6.
4. Grip notches: 6X V-notch through Y on the finger's lower edge, 3 wide × 1.6 deep, pitch 6, first at x 30.
5. Horn holes through the plate: ⌀7 centre + 4X ⌀3.2 on ⌀13.86 PCD at 45°.

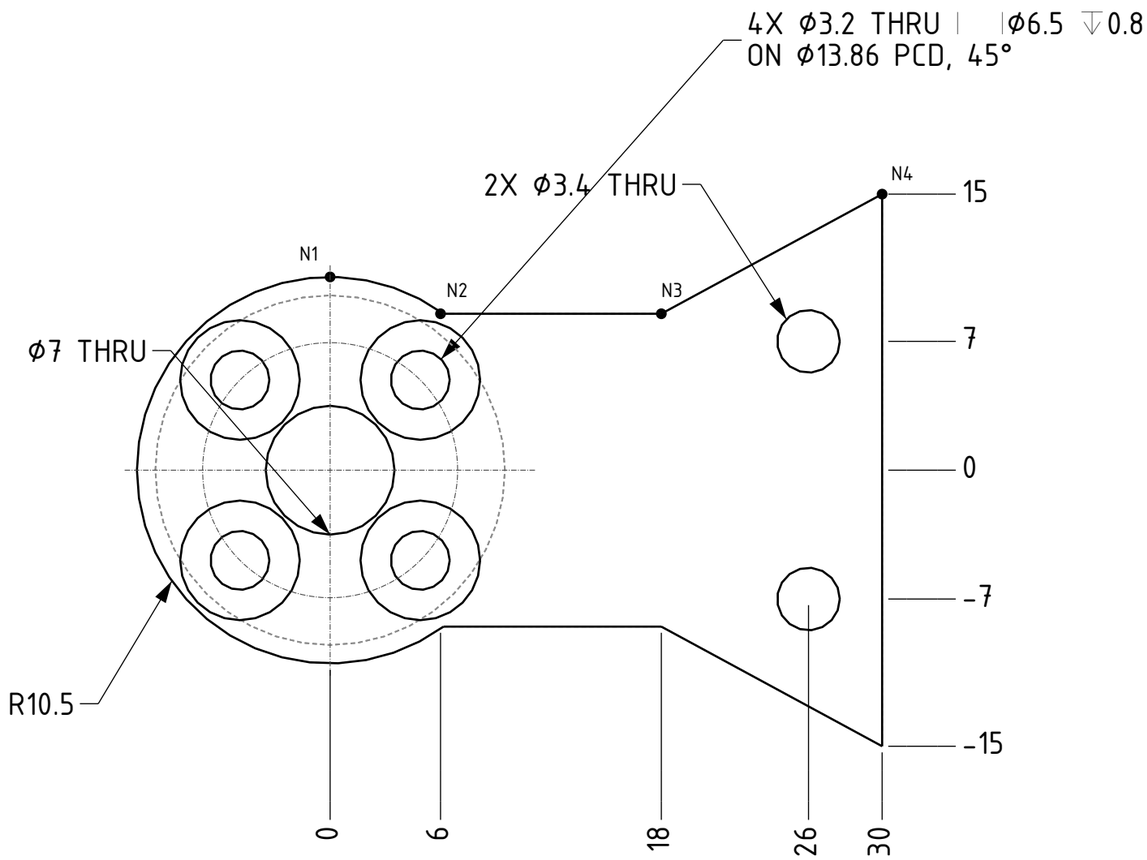
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Bolts to the jaw servo's output horn with 4x M3 x 6; closed position is jaw X along the gripper X.
- Print: horn plate down; support under the finger.

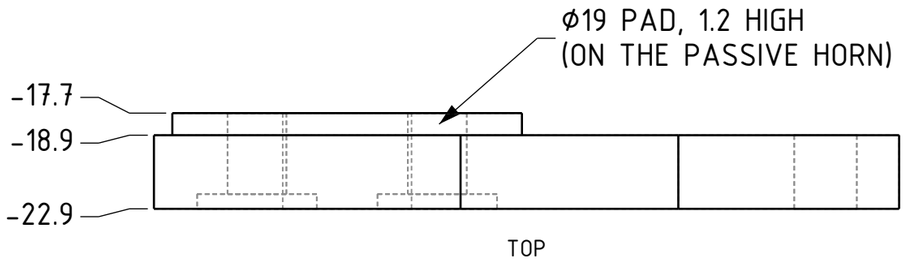
THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates

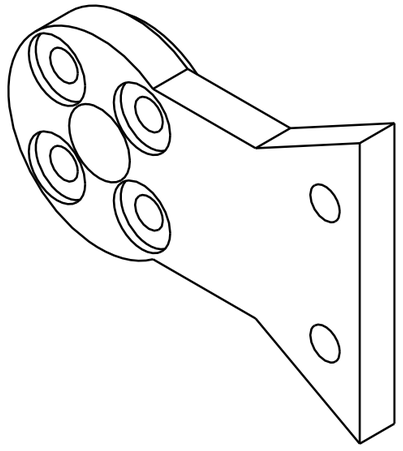
PROJECT TALOS 5-axis arm	EST. MASS 11 g
PART Moving jaw	SCALE 2:1
PART NO. TAL-107	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 18 of 21



FRONT (OUTER FACE, -Y)



TOP



ISOMETRIC (NOT TO SCALE)

NECK POINTS (X, Z), MIRRORED IN Z

PT	X	Z
N1	0	±10.5
N2	6	±8.5
N3	18	±8.5
N4	30	±15

FEATURES (in order)

1. Plate, y -22.9 to -18.9: circle R10.5 at the origin + neck polygon N1 to N4 (mirrored in Z). Same outline as the idler plate on TAL-002.
2. Pad ⌀19 on the inner face, y -18.9 to -17.7: it bears on the servo's passive horn.
3. Centre hole ⌀7 through: clears the servo's idler boss (nothing screws into the boss).
4. Horn holes: 4X ⌀3.2 through on ⌀13.86 PCD at 45°, counterbore ⌀6.5 0.8 deep from the outer face (4X M3 × 6 into the passive horn).
5. Tab holes: 2X ⌀3.4 through at x 26, z ±7 (M3 × 10 into the arm link).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- One on each arm link (TAL-103, TAL-104), in the pocket on the link's -Y side: 2X M3 × 10 into the link, 4X M3 × 6 into the servo's passive horn, which turns on the idler boss. No hub cap on this side: one wide enough for the screws would hit the parent servo's plugs.
- Print: outer face down; no support.

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates

PROJECT TALOS 5-axis arm	EST. MASS 4 g
PART Idler plate	SCALE 3:1
PART NO. TAL-110	UNITS / REV mm / A
QTY / ARM 2	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 20 of 21